

- Ask the user to pick a provider.
- Find the best-match provider based on a set of criteria.

If you want the user to pick a provider, calling `getProviders()` on the `LocationManager` will give you a List of providers, which you can then present to the user for selection. Alternatively, you can create and populate a `Criteria` object, stating the particulars of what you want out of a `LocationProvider`. Here are some of the criteria you can specify:

- `setAltitudeRequired()` : Indicates whether or not you need the current altitude
- `setAccuracy()` : Sets a minimum level of accuracy, in metres, for the position
- `setCostAllowed()` : Controls if the provider must be free or if it can incur a cost on behalf of the device user

Given a filled-in `Criteria` object, call `getBestProvider()` on your `LocationManager`. Android will sift through the criteria and give you the best answer. Note that not all of your criteria may be met. All, but the monetary cost criterion, might be relaxed if nothing matches.

You're also welcome to hardwire in a `LocationProvider` name (e.g., `GPS_PROVIDER`), perhaps just for testing purposes.

Once you know the name of the `LocationProvider`, you can call `getLastKnownPosition()` to find out where you were recently. Note that "recently" might be fairly out of date (e.g., the phone was turned off) or even null if there has been no location recorded for that provider yet. Calling `getLastKnownPosition()` incurs no monetary or power cost, since the provider doesn't need to be activated to get the value.

This method returns a `Location` object, which can give you the latitude and longitude of the device in degrees as a Java `double`. If the particular location provider offers other data, you can get that as well:

- For altitude, `hasAltitude()` will tell you if there's an altitude value, and `getAltitude()` will return the altitude in metres.
- For bearing (i.e., compass-style direction), `hasBearing()` will tell you if there's a bearing available, and `getBearing()` will return it as degrees to the East of true North.
- For speed, `hasSpeed()` will tell you if the speed is known and `getSpeed()` will return the speed in metres per second.

4.4 On the Move

Not all location providers are necessarily immediately responsive. GPS, for example, requires activating a radio and getting a fix from the